3-27-23

$$A^*(b-\hat{b}) = 0$$

$$A^*\hat{b} = A^*b \qquad \hat{b} \in R(A)$$

$$\boxed{A^*Ax = A^*b} \qquad \hat{b} = Ax$$

Thm: (Fredholm alternative): $Ax = b$ has a solution iff $\langle b, v\rangle = 0$ for every $v \in N(A^*)$.

$$b \in R(A) \iff b \perp N(A^*)$$

Ex: $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $\qquad A^* = A^T = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$

$R(A) = \text{span}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$ $\qquad N(A) = \text{span}\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right)$

$R(A^*) = \text{span}\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right)$ $\qquad N(A^*) = \text{span}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$

$\dim(R(A)) = \dim(R(A^T))$

Def: The rank of a matrix $= \dim R(A) = \dim R(A^T)$

i.e., # of linearly indep. cols.

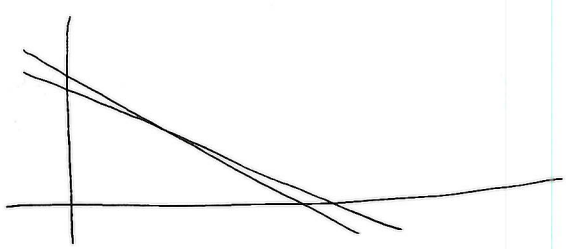
# of linearly indep. rows.

If A is $m \times n$ matrix of rank r:

$\dim(A) = r$	$\dim(A^T) = r$
$\dim N(A) = n-r$	$\dim(N(A^T)) = m-r$

Start

I) Matrix Condition Number



$$\begin{bmatrix} 1+3\epsilon & 1-3\epsilon \\ 3-\epsilon & 3+\epsilon \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

- if $\epsilon = 0$ then the matrix becomes linearly dependent
- if ϵ is small, then we have a poorly cond. conditioned matrix.

A) Derivation

① Want to solve: $Ax = b$. $\quad x_0 = A^{-1}b$.

② What we actually have

$$(A + \epsilon E)\underline{x} = b$$

How does x compare from ① as compared to ②?

$$x_\epsilon = (A + \epsilon E)^{-1}\underline{b}$$

$$x_\epsilon = [A(I + \epsilon A^{-1}E)]^{-1}\underline{b}$$

$$x_\epsilon = [I + \epsilon A^{-1}E]^{-1}A^{-1}b$$

Assume this norm is less than 1

$$\|\epsilon A^{-1}E\| < 1$$

True Solution $\quad x_\epsilon = [I - \epsilon A^{-1}E]\underline{x}_0 + O(\|\epsilon A^{-1}E\|^2)$

$$x_\epsilon = \underline{x}_0 - \epsilon A^{-1}E\underline{x}_0$$

Perterbation on the true solution

Use here

relative error: $\dfrac{\|x_\epsilon - x_0\|}{\|x_0\|} = \dfrac{\|\epsilon A^{-1}E x_0\|}{\|x_0\|} \le \underbrace{\|A^{-1}\|\,\epsilon\,\|E\|}_{\text{All multiplied together}}$

$$= \frac{\|A^{-1}\|\,\|A\|}{\|A\|}\,\epsilon\,\|E\|$$

Rewrite with $\dfrac{\|A\|}{\|A\|}$

Condition number of A

$$\boxed{k(A) = \|A\| \cdot \|A^{-1}\|}$$

= condition number of A.

If A is not invertible, we take $k(A) = \infty$.

Put it all back together

$$\frac{\|x_\epsilon - x_0\|}{\|x_0\|} \le k(A)\,\epsilon\,\frac{\|E\|}{\|A\|}$$

$$\rho = \epsilon\frac{\|E\|}{\|A\|}$$

= relative error in perturbation.

$$\boxed{\frac{\|x_\epsilon - x_0\|}{\|x_0\|} \le k(A)\rho}$$

So this is the condition number multiplied by the relative error in the perturbation

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B) Computing Condition Number

1) Matrix Norm

$$M = \|A\| = \max_{x \ne 0} \frac{\|Ax\|}{\|x\|}$$

— largest vector with A

$$m = \min_{x \ne 0} \frac{\|Ax\|}{\|x\|}$$

$$\|A^{-1}\| = \frac{1}{m}$$

— the smallest vector with A

$$\|A^{-1}\|^{-1} = m$$

2) Compute ~~Matrix~~ Condition

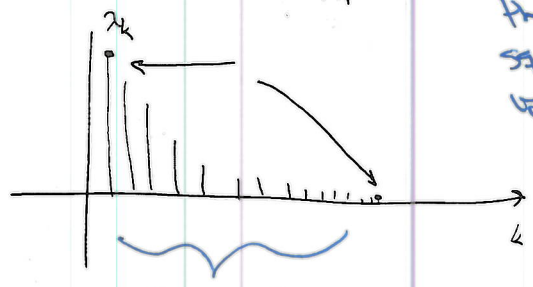
$$\boxed{k(A) = \frac{M}{m}}$$

for an $\|\ \|_2$:

λ: eigen value

$$k(A) = \frac{\sqrt{\lambda_{max}}}{\sqrt{\lambda_{min}}}$$

For L_2 norm the condition number is the ratio of the start of the eigen values.



If you have a large difference of largest and smallest eigen value, then you probably have a poorly conditioned matrix

3)

Rule of thumb:

$$p = \log_{10}(k(A))$$

P is roughly the decimal precision lost if this matrix is computed.

$$\frac{1}{k(A)} + 1 = 1$$

Matrix is bad.

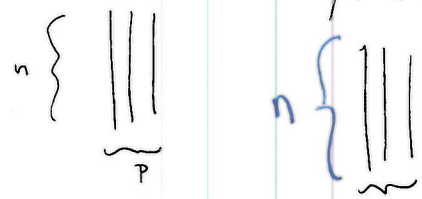
Another little test. We want this value to be greater than 1.

Matlab: cond, rcond

II) Inverse of a small rank adjustment.

A) Small Rank Matrix

X : $n \times p$ $\qquad$ Y : $n \times p$



$p < n$

Multiply the matrices together

XY^H : $n \times n$ $\qquad$ $\text{rank}(XY^H) = p$

But the rank is going to be much smaller rank verses the size

B) Small Rank Adjustment

Take $p=1$

$X = n \times 1$ $\qquad$ $Y = n \times 1$

just some $n \times n$ matrix

A: $n \times n$. Assume that A^{-1} is known.

This is the small rank adjustment $\rightarrow$ Now form $B = A + xy^H$

Question: Can we efficiently find B^{-1}?

Yes!

$$\boxed{B^{-1} = (A + xy^H)^{-1} = A^{-1} - \frac{A^{-1} x y^H A^{-1}}{1 + y^H A^{-1} x}}$$

Matrix Inversion Lemma
Sherman Morrison Formula.

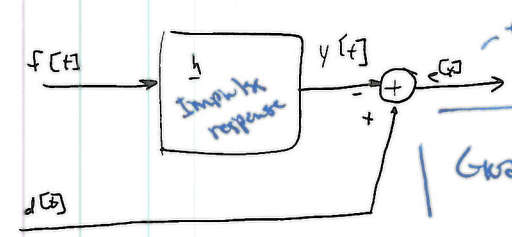
Some names of this bad boy

B) Application

Application: Recursive Least Squares Adaptive Filter. (RLS)

input signal — $f[t]$

desired signal — $d[t]$



error

$e[t] = d[t] - y[t]$

Goal: minimize $\sum_{i=1}^{n} |e[i]|^2$

$y[t]$ is the convolution of the input and impulse response — $y[t] = \sum_{l=0}^{m} f[t-l]\, h[l]$

OH BOY A GRAMMIAN

$$\underbrace{\begin{bmatrix} f[1] & f[-1] & \cdots & f[1-m] \\ f[2] & f[1] & \cdots & f[2-m] \\ & \ddots & \ddots & \\ f[n] & \cdots & & f[n-m] \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[m] \end{bmatrix}}_{h} = \underbrace{\begin{bmatrix} d[1] \\ d[2] \\ \vdots \\ d[n] \end{bmatrix}}_{d} + \begin{bmatrix} e[1] \\ e[2] \\ \vdots \\ e[n] \end{bmatrix}$$

Toeplitz Structure

Bad way

$$(A^H A)^{-1} h = A^H d$$

$$\boxed{h = (A^H A)^{-1} A^H d}$$

WOW! We can use this like we have. But this is brute force

But now lets assume this is running and we get a new row on the matrix A.

Lets use our matrix inversion lemma!

$$A = \begin{bmatrix} f[1] & \cdots & f[1-m] \\ \vdots & & \\ f[N] & \cdots & f[N-m] \end{bmatrix}$$

Think in terms of rows

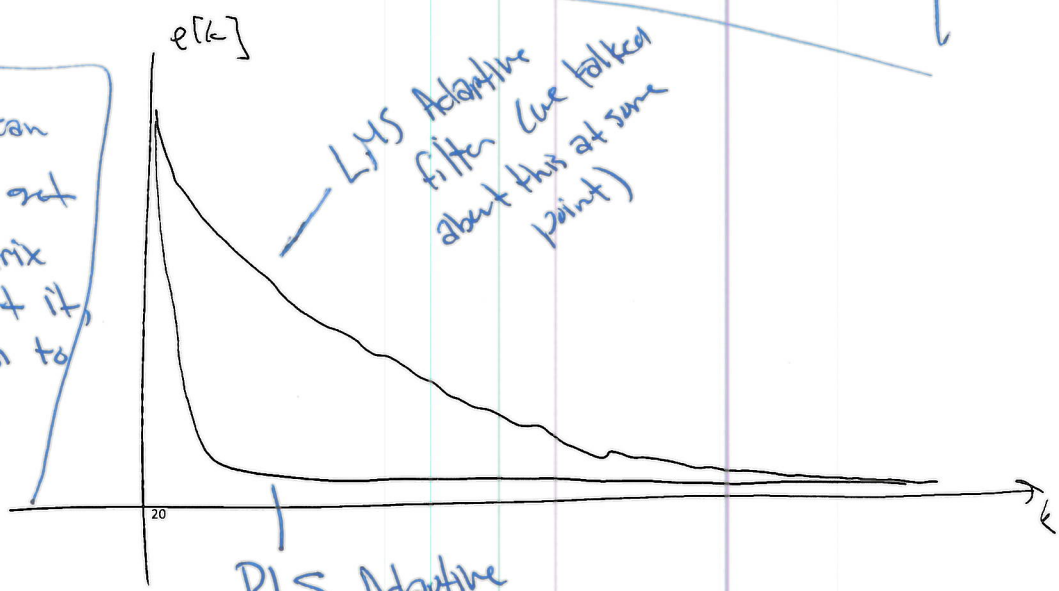
$$= \begin{bmatrix} \text{---} & q[1]^H & \text{---} \\ \text{---} & q[2]^H & \text{---} \\ & \vdots & \\ \text{---} & q[N]^H & \text{---} \end{bmatrix}$$

$$A^H = \begin{bmatrix} | & | & & | \\ q[1] & q[2] & \cdots & q[N] \\ | & | & & | \end{bmatrix}$$

$$(A^H A) = \sum_{i=1}^{N} q[i]\, q[i]^H$$

Use this idea and append the new elements

Add one more row:

$$(A^H A) = \sum_{i=1}^{N+1} q[i]\, q[i]^H = \underbrace{\sum_{i=1}^{N} q[i]\, q[i]^H}_{\text{old } A} + \underbrace{q[N+1]\, q[N+1]^H}_{\text{a rank-one adjustment.}}$$

Because we have $h = (A^H A)^{-1} A^H d$ we can use the method above to get $A^H A$, then use the matrix inversion lemma to invert it, then plug it all back in to get our coefficients h.